1.Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Divide 192.168.10.0/24 into 4 Subnets

To divide this network into 4 equal subnets, we use the /26 subnet mask.

Subnet mask: `255.255.255.192`

Subnet	Network Address	Broadcast Address	Subnet Mask
1	192.168.10.0	192.168.10.63	255.255.255.192
2	192.168.10.64	192.168.10.127	255.255.255.192
3	192.168.10.128	192.168.10.191	255.255.255.192
4	192.168.10.192	192.168.10.255	255.255.255.192

**2.For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.**

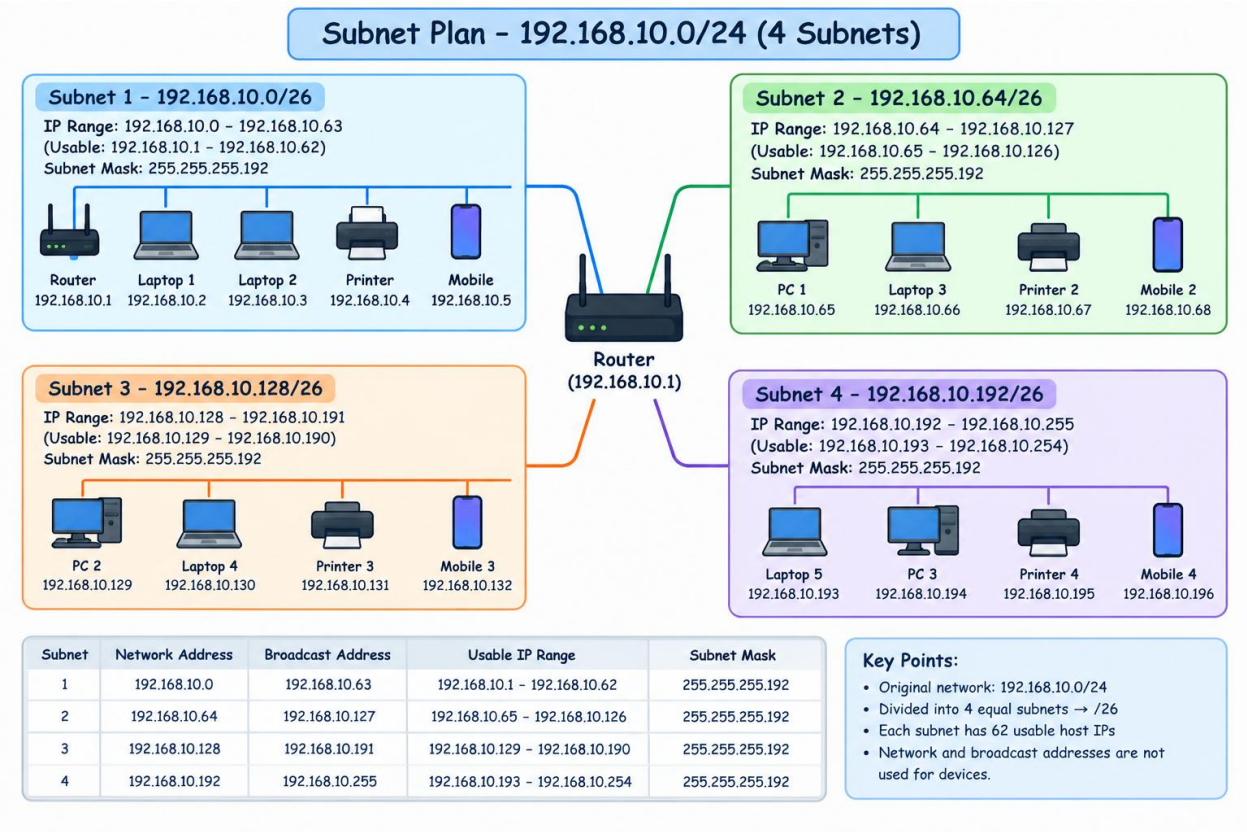
IP Address Assignment for 5 Devices

I will use Subnet 1: 192.168.10.0/26.

- Network Address: `192.168.10.0`
- Broadcast Address: `192.168.10.63`
- Subnet Mask: `255.255.255.192`
- Valid Host Range: `192.168.10.1 – 192.168.10.62`

Device	Assigned IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile Phone	192.168.10.5

3.Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



Subnet	Network Address	Broadcast Address	Usable IP Range	Subnet Mask
1	192.168.10.0	192.168.10.63	192.168.10.1 – 192.168.10.62	255.255.255.192
2	192.168.10.64	192.168.10.127	192.168.10.65 – 192.168.10.126	255.255.255.192
3	192.168.10.128	192.168.10.191	192.168.10.129 – 192.168.10.190	255.255.255.192
4	192.168.10.192	192.168.10.255	192.168.10.193 – 192.168.10.254	255.255.255.192

4.Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step-by-Step Subnetting (Easy Language)

Step 1: Find the subnet mask

My original network is `192.168.10.0/24`. I need to divide it into 4 equal subnets.

- New subnet mask: `/26`
- Subnet mask: `255.255.255.192`

Step 2: Calculate the number of hosts

- Each subnet has 64 IP addresses.

- 2 addresses are reserved for the network and broadcast.
- So, each subnet has 62 usable IP addresses.

Step 3: Divide the network into 4 subnets

Subnet	Network Address	Broadcast Address
1	192.168.10.0	192.168.10.63
2	192.168.10.64	192.168.10.127
3	192.168.10.128	192.168.10.191
4	192.168.10.192	192.168.10.255

Step 4: Assign IP addresses to devices

I chose Subnet 1 for my devices.

Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

Step 5: Check the addresses

All devices have different IP addresses. None uses the network address or broadcast address.